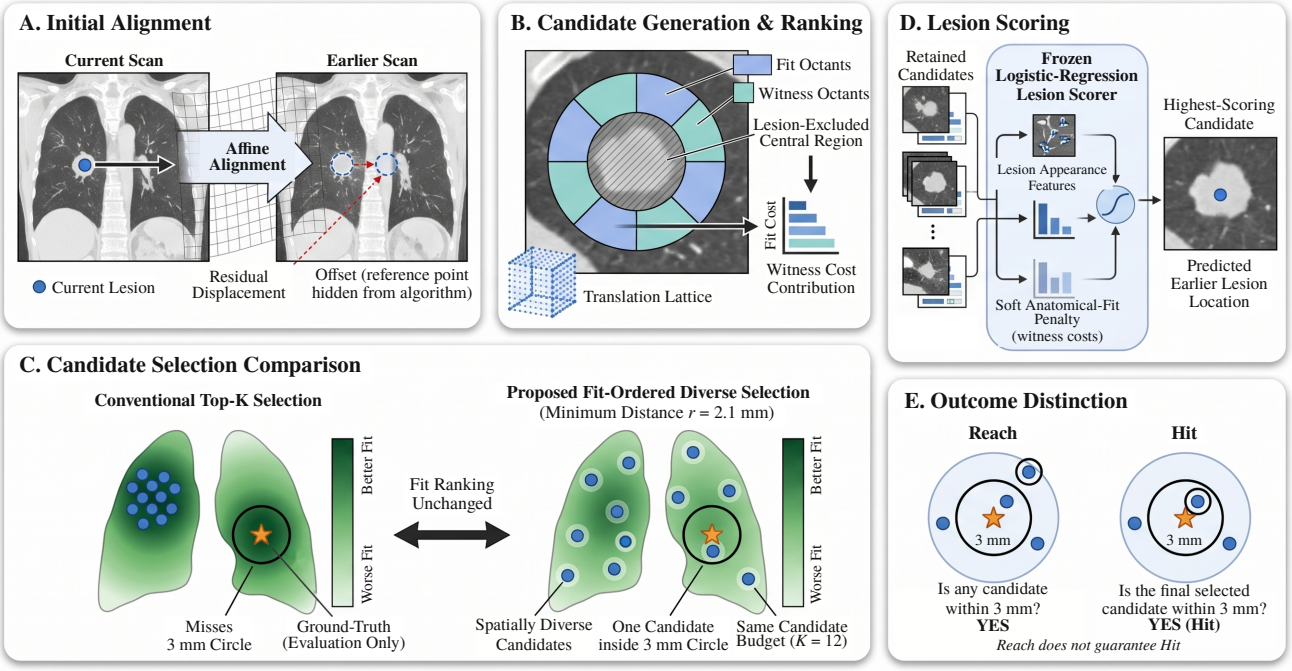

A. Initial Alignment
Current Scan
Earlier Scan
Affine Alignment
Current Lesion
Residual Displacement
Offset (reference point hidden from algorithm)
B. Candidate Generation & Ranking
Fit Octants
Witness Octants
Lesion-Excluded Central Region
Fit Cost
Witness Cost Contribution
Translation Lattice
D. Lesion Scoring
Retained Candidates
Frozen Logistic-Regression Lesion Scorer
Lesion Appearance Features
Soft Anatomical-Fit Penalty (witness costs)
Highest-Scoring Candidate
Predicted Earlier Lesion Location
C. Candidate Selection Comparison
Conventional Top-K Selection
Better Fit
Worse Fit
Misses 3 mm Circle
Ground-Truth (Evaluation Only)
Fit Ranking Unchanged
Proposed Fit-Ordered Diverse Selection
(Minimum Distance r = 2.1 mm)
Spatially Diverse Candidates
One Candidate inside 3 mm Circle
Same Candidate Budget (K = 12)
E. Outcome Distinction
Reach
Hit
3 mm
Is any candidate within 3 mm?
YES
Is the final selected candidate within 3 mm?
YES (Hit)
Reach does not guarantee Hit